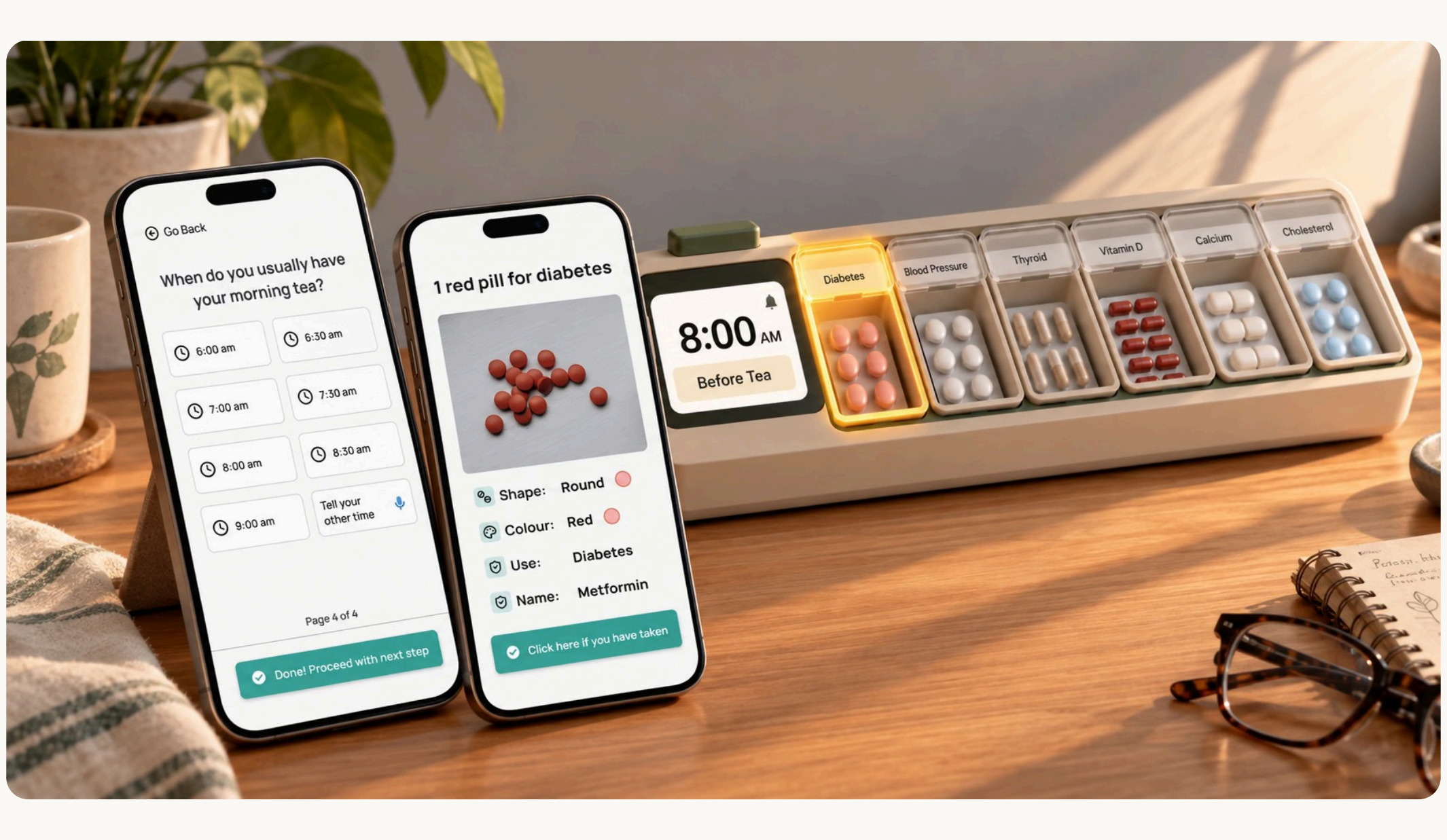
Simplifying Polypharmacy for Older Adults

Designing an accessible experience for managing multiple medications and everyday care.

My Role
Ux Research, Use interviews, Behavioural Research, Interaction design, Information architecture, Visual design.

Project Type
UX Research, Service design, application design, Product design

Tools Used



What does the current scenario look like?

1 **Managing around 4 to 12 medicines each day**
Managing care can mean multiple medicines, frequent doctor visits, and changing prescriptions.

2 **Medical Information Stored Separately**
Medicine details and medical records are often scattered, making quick access difficult.

3 **Multiple Doctor's visit and Prescription change**
Frequent doctor visits lead to changing prescriptions and make care harder to manage.

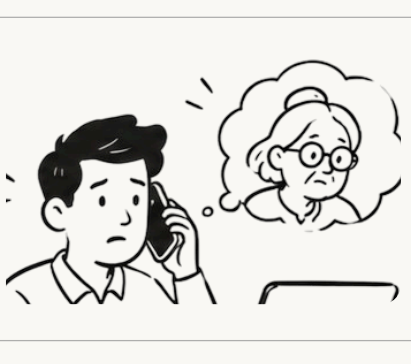
4 **Tedious Pill Arrangement**
Sorting medicines into pill boxes by day and time is repetitive and time-consuming.

Qualitative Research

Interviewed 7elderly people, 9 family members, 1 medical student

"When my routine changes, I sometimes forget to take them, which affects my health."

"I remember my medicines through routines, like after brushing or before breakfast."



"I do not want to depend on my family all the time"

" Medical records are scattered"

"Sometimes we need a Caregiver, but it is hard to find them"

"I want to use digital tools, but they're often difficult to understand and use."

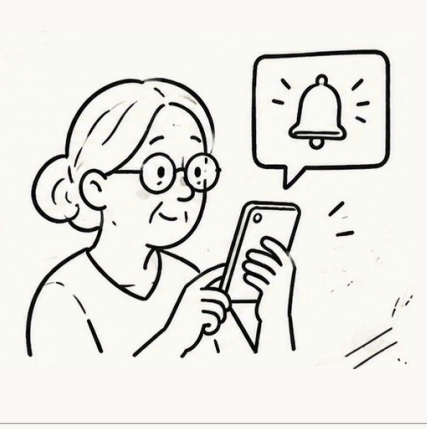
Quantitative Research

Surveyed 102 people

70%
"Still rely on memory"

65%
"Store medical records phscially"

52%
"See value in medicine remainders"



Secondary Research

Medicines are remembered by how they look
They often recognise medicines by their shape, size, colour or where they keep it.

Medicines are connected to routines
They often remember medicines as before tea, after lunch...

Remainders helpful when the medicines are new
Helpful until they get used to it

A digital solution needs to be genuinely useful
If their existing solution is working for them elready, they want something more from a digital solution

Elderly, wanted more features
Like medication schedules, reminders, dose tracking, refill alerts, and medicine information in one place.

Problem Statement

"How might we simplify medication and everyday care while keeping families involved?"

User personas

01 ELDERLY PERSON

Meena Joshi · 68
Retired teacher · Lives with her husband

Meena takes multiple medicines every day. She has developed her own way of remembering them through their appearance and her daily routine.

Needs
- Easily recognise the right medicine
- Store medical history and records
- Add medicines and records easily
- Get reminders when needed
- Adaptable remainder for routine change

Frustrations
- Similar-looking medicines get confusing
- Changing prescriptions disturb her routine
- Too many steps or information can feel overwhelming.

02 FAMILY CAREGIVER

Rahul Joshi · 39
Working professional · Meena's son

Rahul helps manage her mothers healthcare, but work means he cannot always be around to check medicines, appointments or accompany her

Needs
- Know whether medicines are taken
- Quickly add or Update medicines
- Keep records and appointments together
- Find trustable caregiver
- Adaptable remainder for routine change

Frustrations
- Managing many medicines takes time
- Medical information is scattered
- Hard to know what happened when he wasn't there

The solution

Grouped medicine system
Medicines taken around the same routine are grouped togeth

More personal routine remainders
Remainders can be set around smaller everyday routines like before tea, after tea...

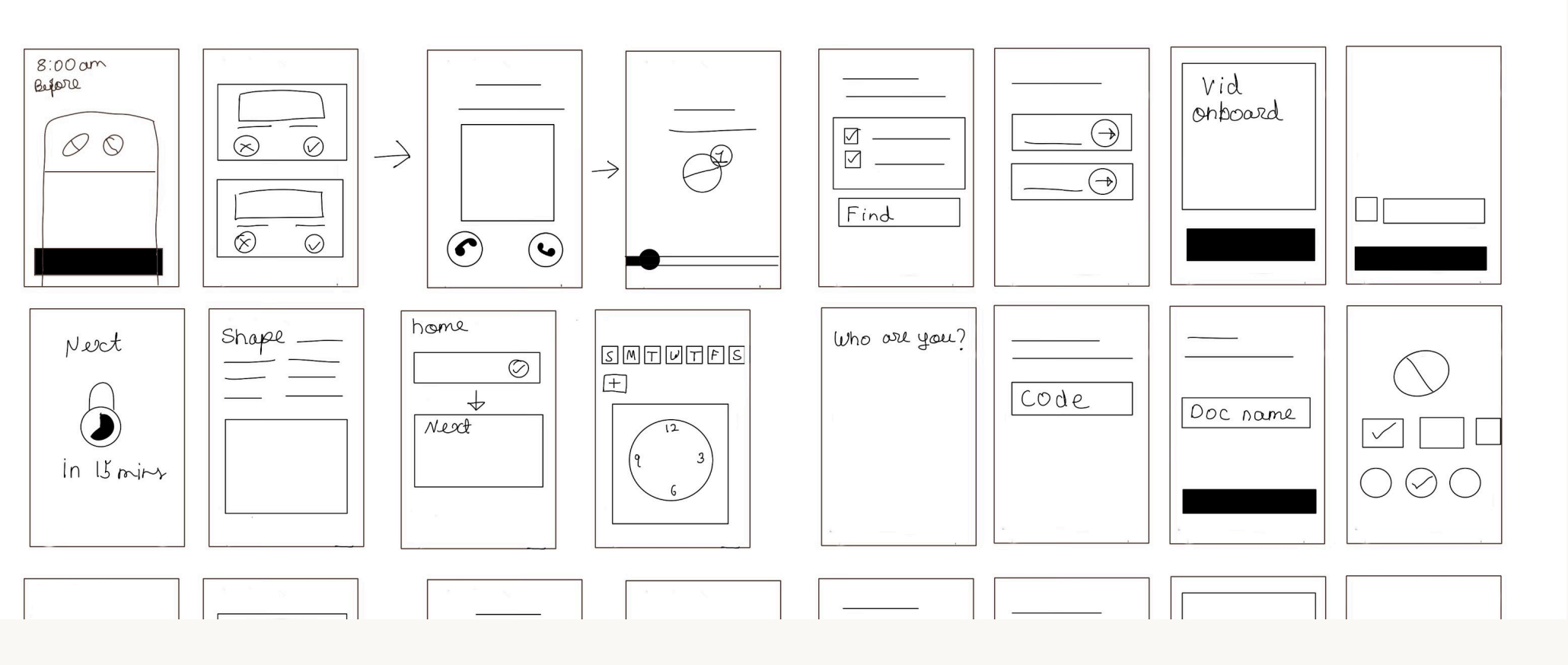
Easy to recognise medicine remainders
Remainders that show the medicine's image, colour, shape and familiar name

Easy medicine management
Medicines can be added quickly, updated, or turned off when no longer needed

Family medicine tracking
Family members can see which medicines were taken or missed and keep track of the medicines

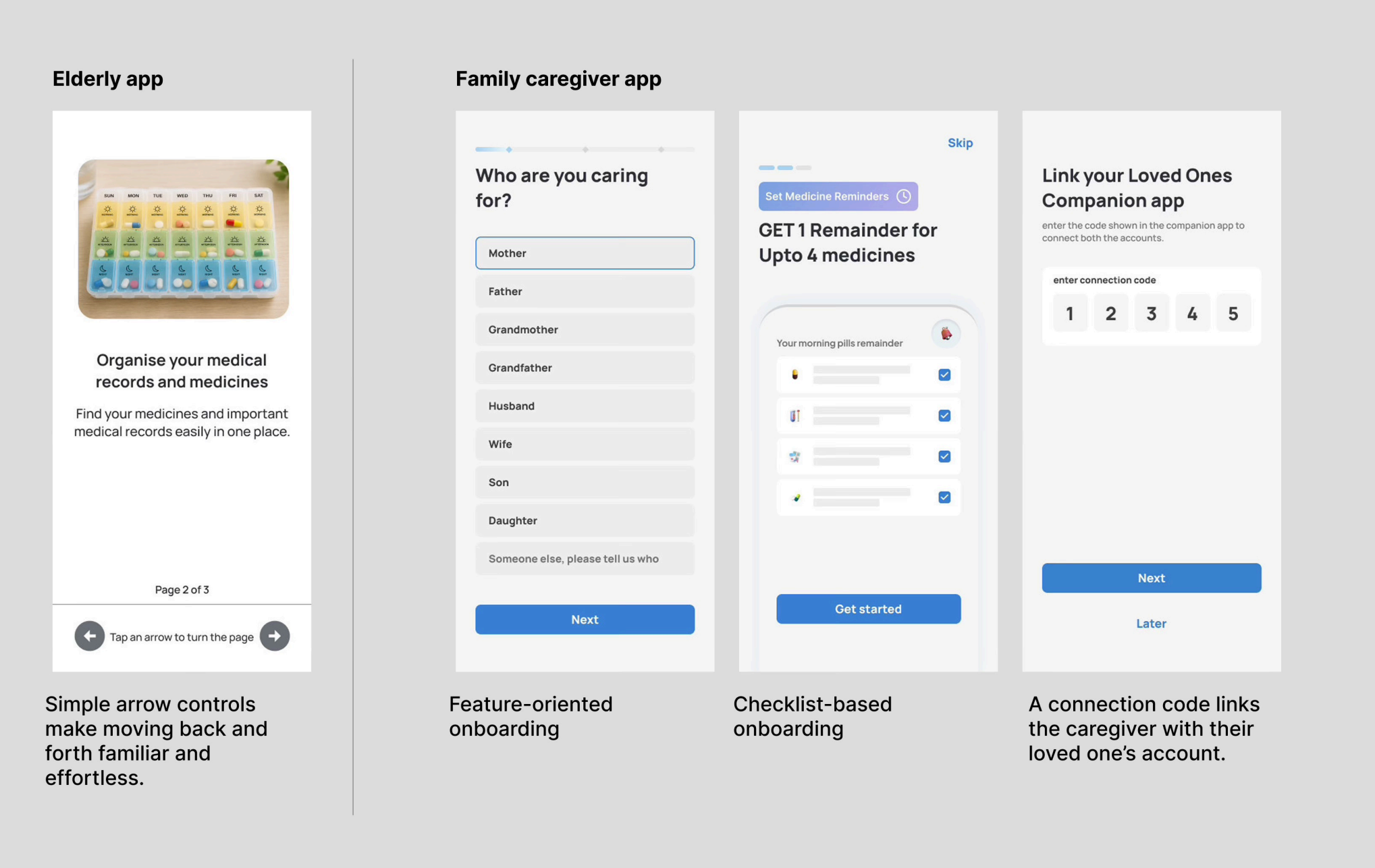
Book caregiver for specific needs
Families can find caregiver for specific needs and timeline

Wireframes



High-fidelity screens

Onboarding



Elderly app

Organise your medical records and medicines
Find your medicines and important medical records easily in one place.
Page 2 of 3
Tap an arrow to turn the page

Simple arrow controls make moving back and forth familiar and effortless.

Family caregiver app

Who are you caring for?
Mother
Father
Grandmother
Grandfather
Husband
Wife
Son
Daughter
Someone else, please tell us who
Next

Feature-oriented onboarding

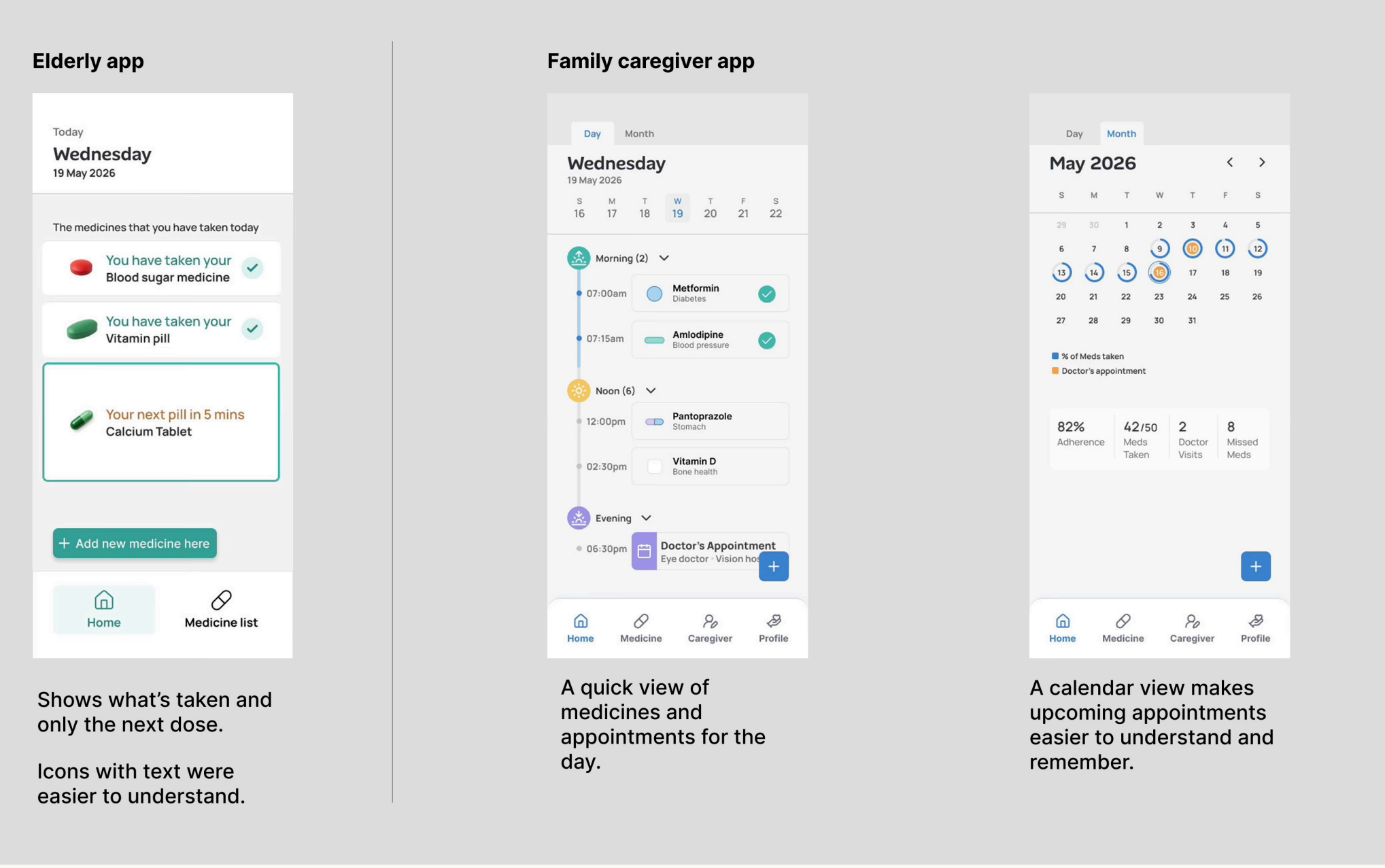
Skip
Set Medicine Reminders
GET 1 Remainder for Upto 4 medicines
Your morning pills remainder
Get started

Checklist-based onboarding

Link your Loved Ones Companion app
enter the code shown in the companion app to connect both the accounts.
enter connection code
1 2 3 4 5
Next
Later

A connection code links the caregiver with their loved one's account.

Home screen



Elderly app

Today
Wednesday
19 May 2026
The medicines that you have taken today
You have taken your Blood sugar medicine
You have taken your Vitamin pill
Your next pill in 5 mins Calcium Tablet
Add new medicine here
Home
Medicine list

Shows what's taken and only the next dose.

Icons with text were easier to understand.

Family caregiver app

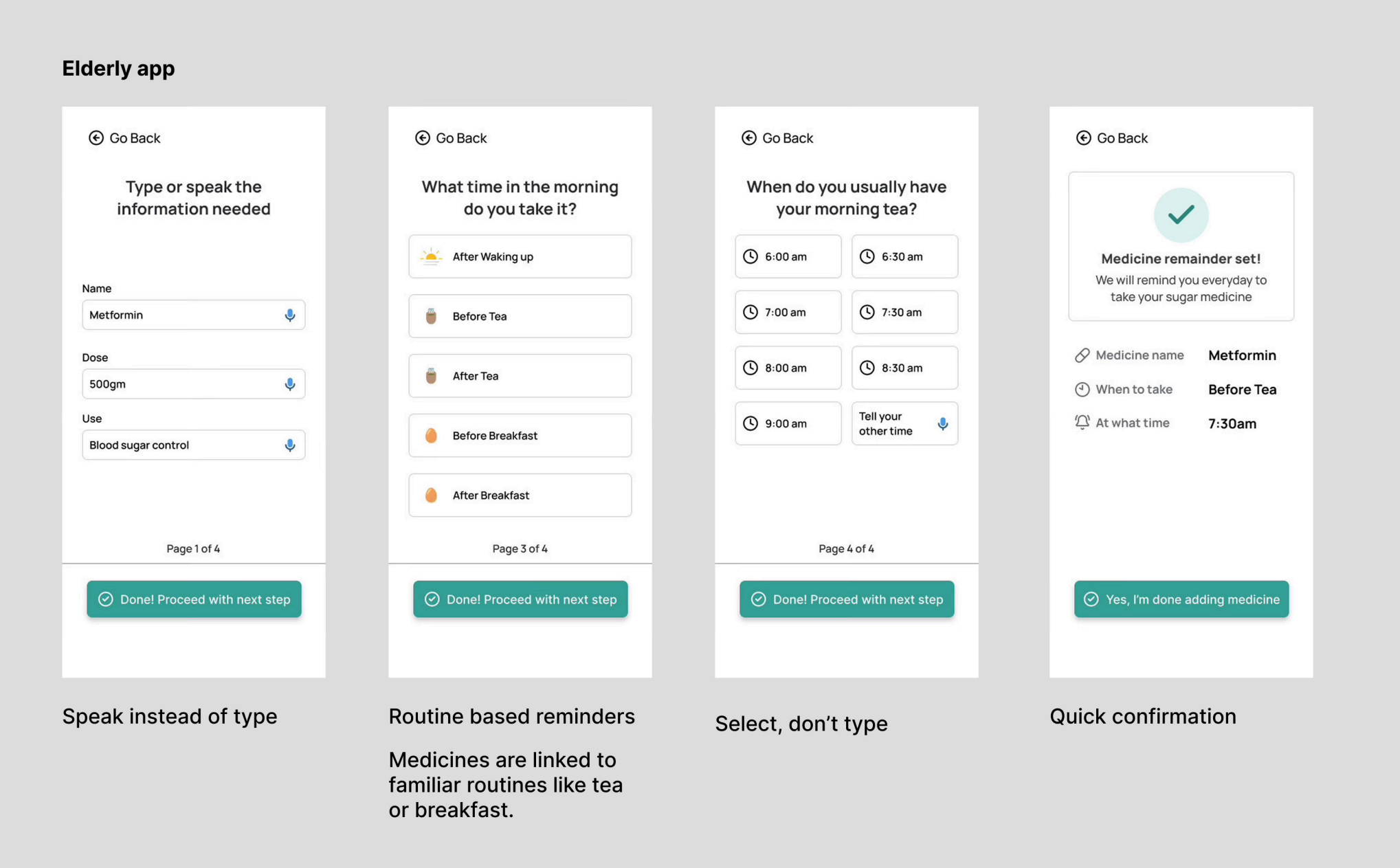
Day Month
Wednesday
19 May 2026
Morning (2)
07:00am Metformin Diabetes
07:15am Amlodipine Blood pressure
Noon (6)
12:00pm Pantoprazole Stomach
02:30pm Vitamin D Bone health
Evening
06:30pm Doctor's Appointment Eye doctor · Vision ho
Home Medicine Caregiver Profile

A quick view of medicines and appointments for the day.

Day Month
May 2026
% of Meds taken
Doctor's appointment
82% Adherence
42/50 Meds Taken
2 Doctor Visits
8 Missed Meds
Home Medicine Caregiver Profile

A calendar view makes upcoming appointments easier to understand and remember.

Adding medicine - Elderly app



Elderly app

Go Back
Type or speak the information needed
Name: Metformin
Dose: 500gm
Use: Blood sugar control
Page 1 of 4
Done! Proceed with next step

Speak instead of type

Go Back
What time in the morning do you take it?
After Waking up
Before Tea
After Tea
Before Breakfast
After Breakfast
Page 3 of 4
Done! Proceed with next step

Routine based reminders

Medicines are linked to familiar routines like tea or breakfast.

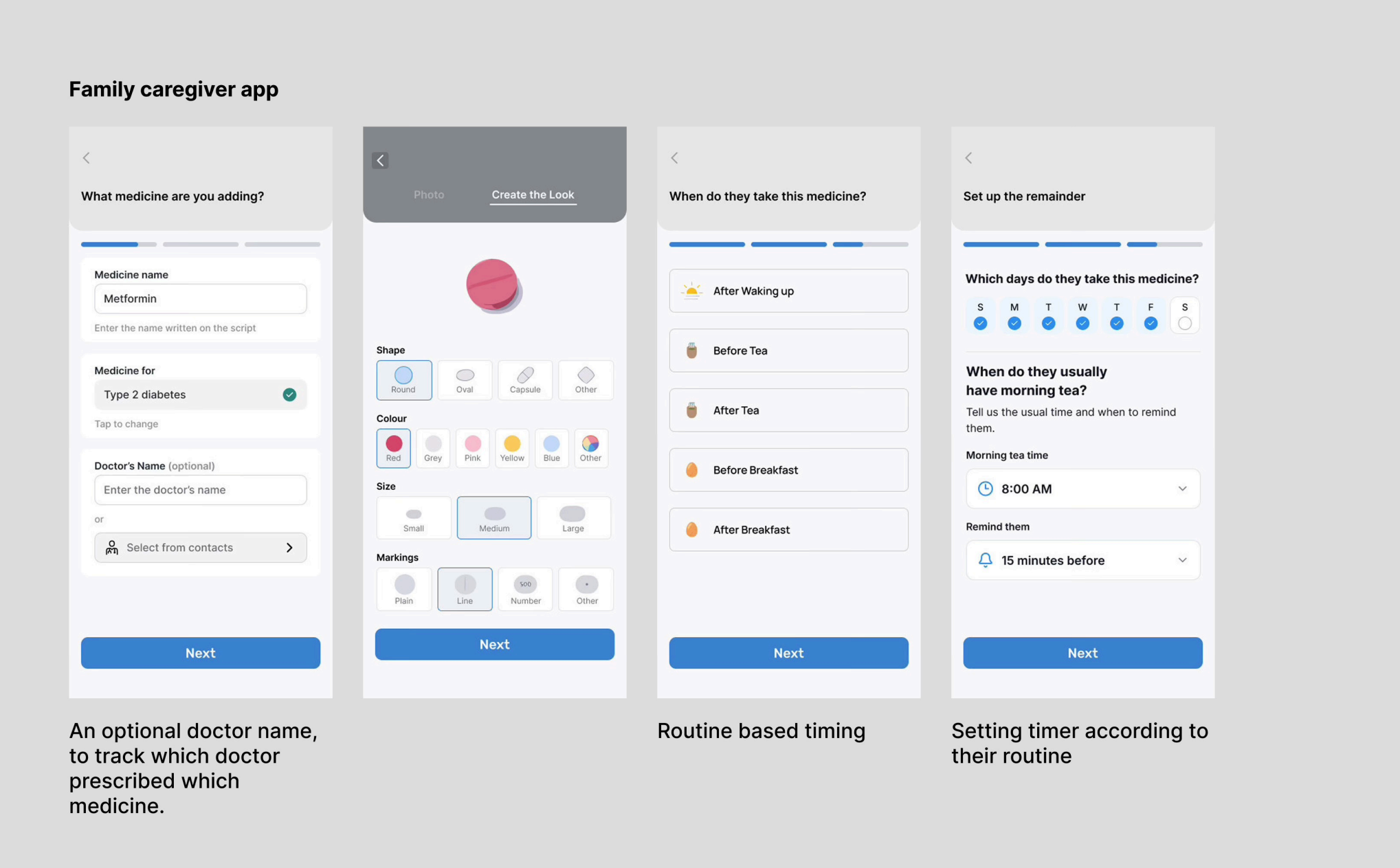
Go Back
When do you usually have your morning tea?
6:00 am
6:30 am
7:00 am
7:30 am
8:00 am
8:30 am
9:00 am
Tell your other time
Page 4 of 4
Done! Proceed with next step

Select, don't type

Go Back
Medicine remainder set!
We will remind you everyday to take your sugar medicine
Medicine name: Metformin
When to take: Before Tea
At what time: 7:30am
Yes, I'm done adding medicine

Quick confirmation

Adding medicine - Family caregiver app



Family caregiver app

What medicine are you adding?
Medicine name: Metformin
Enter the name written on the script
Medicine for: Type 2 diabetes
Tap to change
Doctor's Name (optional)
Enter the doctor's name
or
Select from contacts
Next

An optional doctor name, to track which doctor prescribed which medicine.

Photo Create the Look
Shape: Round Oval Capsule Other
Colour: Red Grey Pink Yellow Blue Other
Size: Small Medium Large
Markings: Plain Line Number Other
Next

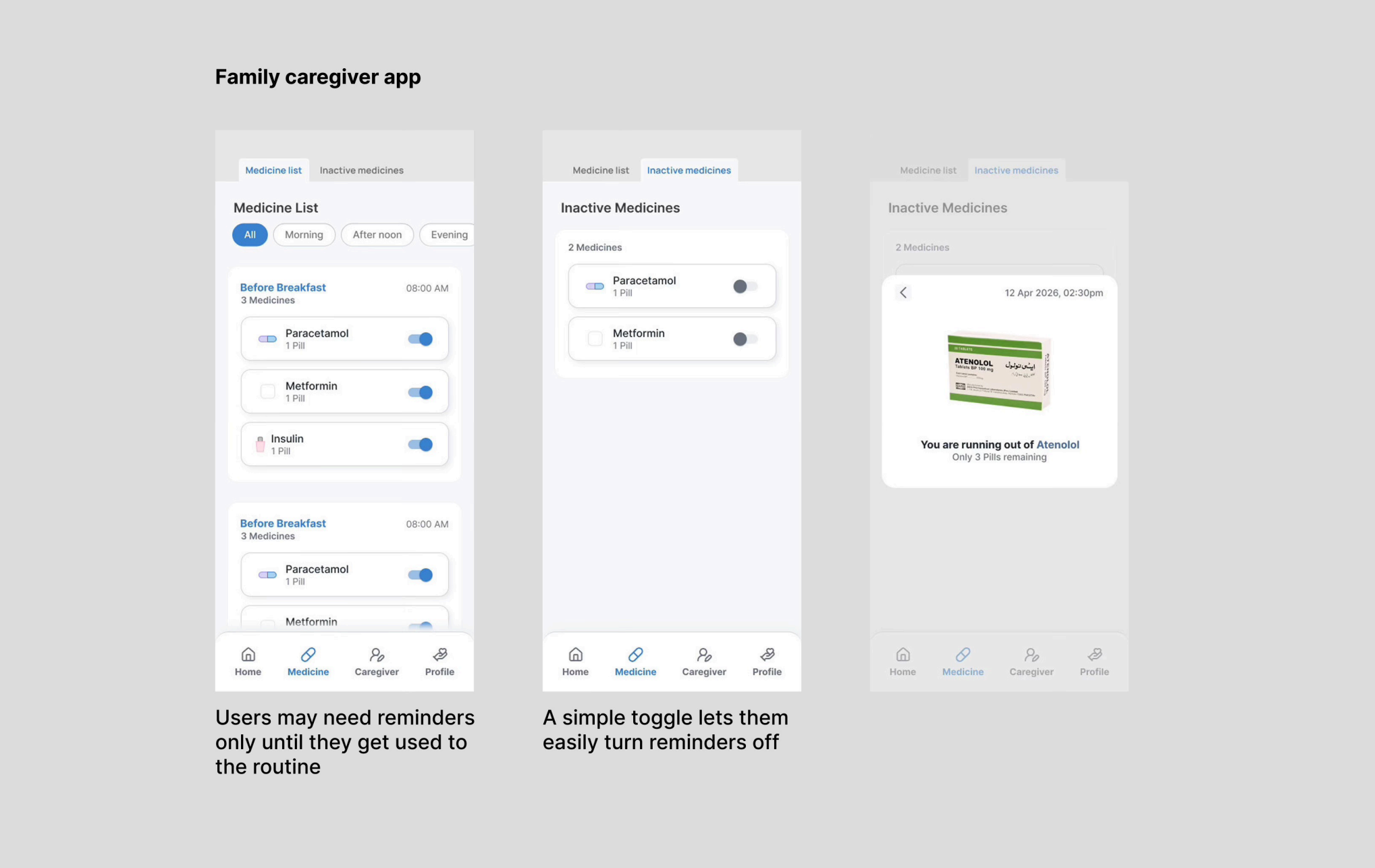
When do they take this medicine?
After Waking up
Before Tea
After Tea
Before Breakfast
After Breakfast
Next

Routine based timing

Set up the remainder
Which days do they take this medicine?
S M T W T F S
When do they usually have morning tea?
Tell us the usual time and when to remind them.
Morning tea time: 8:00 AM
Remind them: 15 minutes before
Next

Setting timer according to their routine

Managing medicine



Family caregiver app

Medicine list Inactive medicines
Medicine List
All Morning After noon Evening
Before Breakfast 3 Medicines 08:00 AM
Paracetamol 1 Pill
Metformin 1 Pill
Insulin 1 Pill
Before Breakfast 3 Medicines 08:00 AM
Paracetamol 1 Pill
Metformin
Home Medicine Caregiver Profile

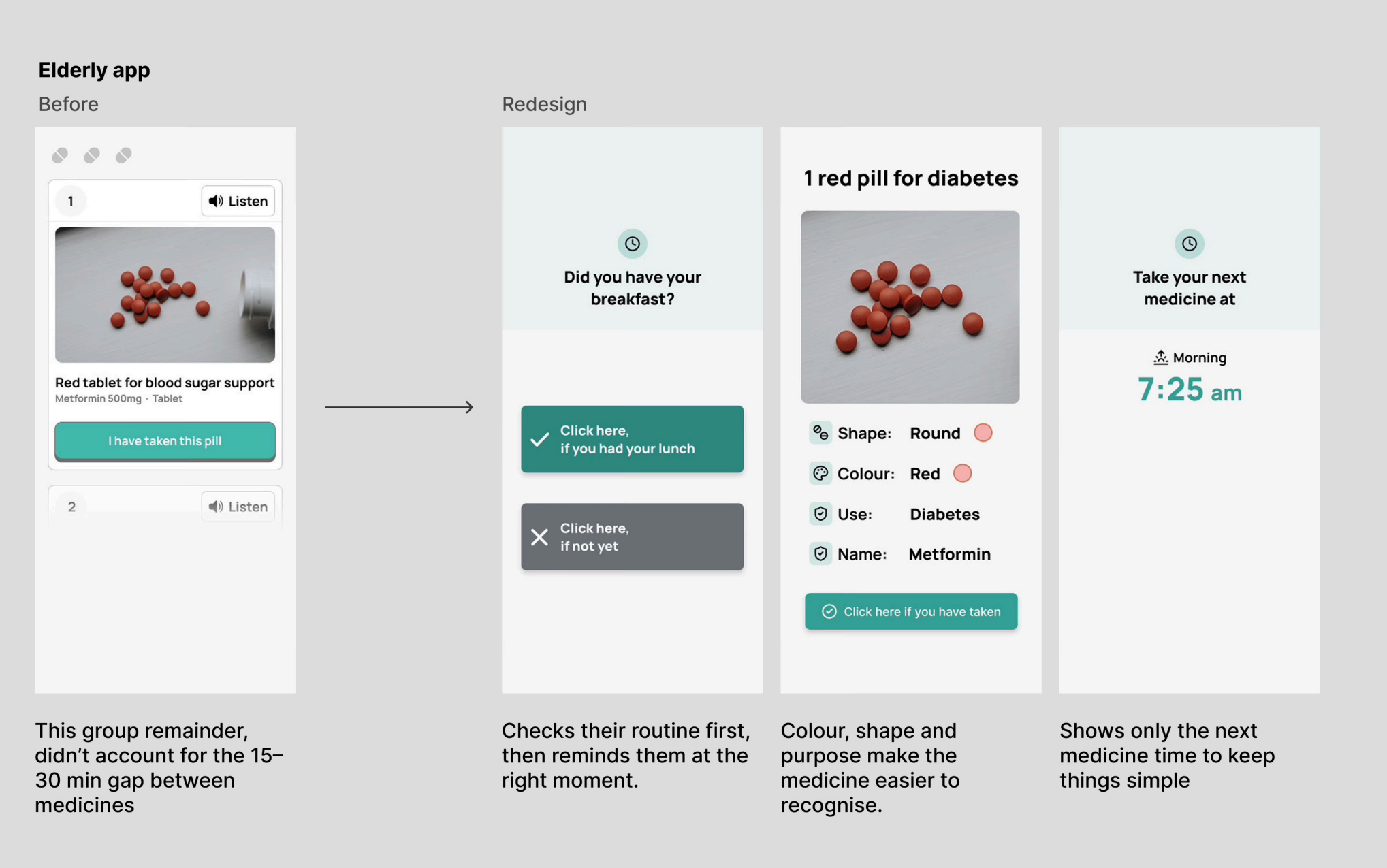
Users may need reminders only until they get used to the routine

Medicine list Inactive medicines
Inactive Medicines
2 Medicines
Paracetamol 1 Pill
Metformin 1 Pill
Home Medicine Caregiver Profile

A simple toggle lets them easily turn reminders off

Inactive Medicines
2 Medicines
12 Apr 2026, 02:30pm
You are running out of Atenolol
Only 3 Pills remaining

Medicine remainders



Elderly app

Before

1 Listen
Red tablet for blood sugar support
Metformin 500mg · Tablet
I have taken this pill
2 Listen

This group remainder, didn't account for the 15–30 min gap between medicines

Redesign

Did you have your breakfast?
Click here, if you had your lunch
Click here, if not yet

Checks their routine first, then reminds them at the right moment.

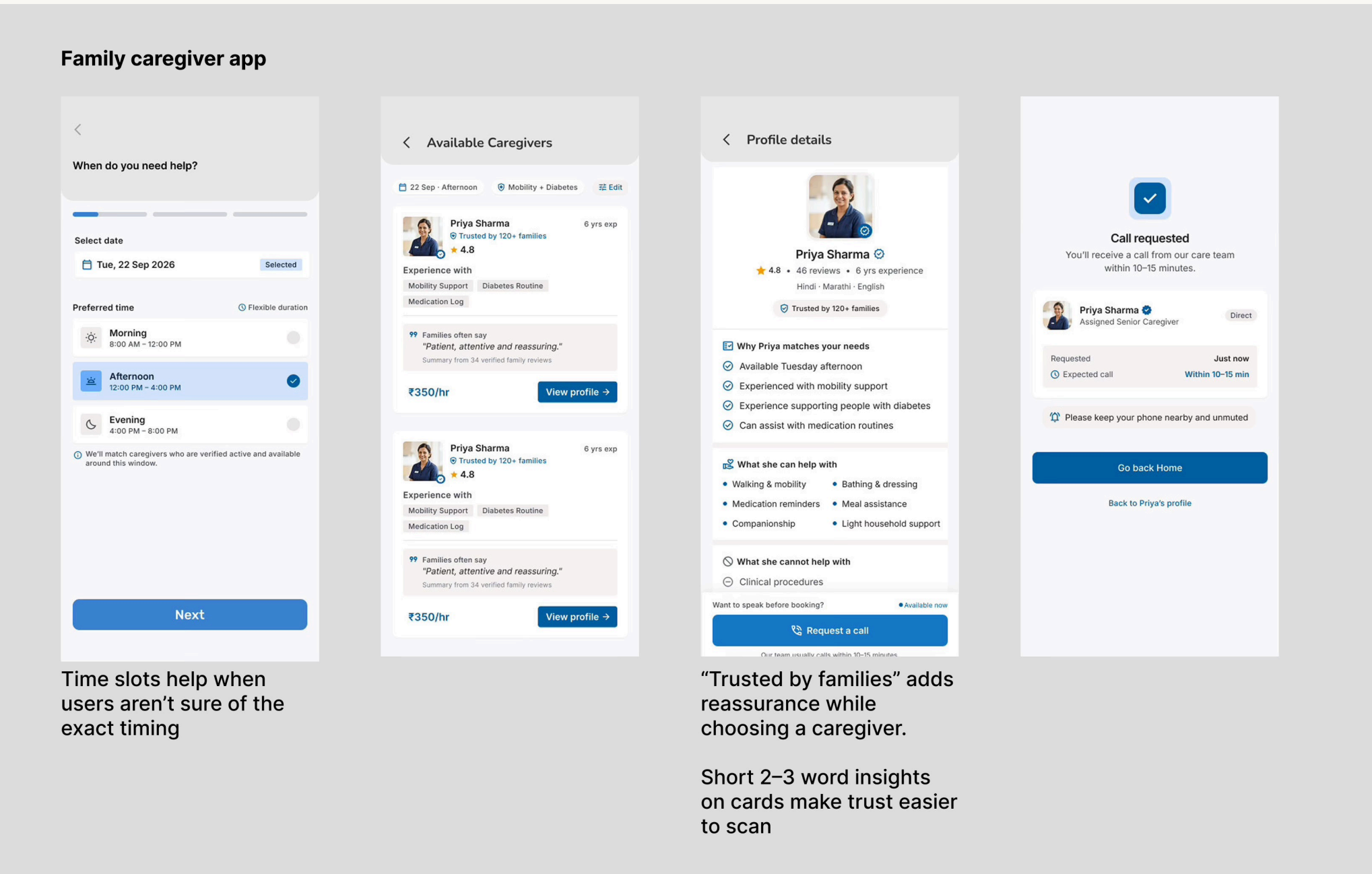
1 red pill for diabetes
Shape: Round
Colour: Red
Use: Diabetes
Name: Metformin
Click here if you have taken

Colour, shape and purpose make the medicine easier to recognise.

Take your next medicine at
Morning
7:25 am

Shows only the next medicine time to keep things simple

Caregiver booking



Family caregiver app

When do you need help?
Select date: Tue, 22 Sep 2026 Selected
Preferred time Flexible duration
Morning 8:00 AM - 12:00 PM
Afternoon 12:00 PM - 4:00 PM
Evening 4:00 PM - 8:00 PM
We'll match caregivers who are verified active and available around this window.
Next

Time slots help when users aren't sure of the exact timing

Available Caregivers
22 Sep · Afternoon Mobility + Diabetes Edit
Priya Sharma 6 yrs exp
Trusted by 120+ families
4.8
Experience with
Mobility Support Diabetes Routine Medication Log
Families often say "Patient, attentive and reassuring."
₹350/hr View profile

Profile details
Priya Sharma
4.8 · 46 reviews · 6 yrs experience
Hindi · Marathi · English
Trusted by 120+ families
Why Priya matches your needs
Available Tuesday afternoon
Experienced with mobility support
Experience supporting people with diabetes
Can assist with medication routines
What she can help with
Walking & mobility
Bathing & dressing
Medication reminders
Meal assistance
Companionship
Light household support
What she cannot help with
Clinical procedures
Want to speak before booking?
Request a call

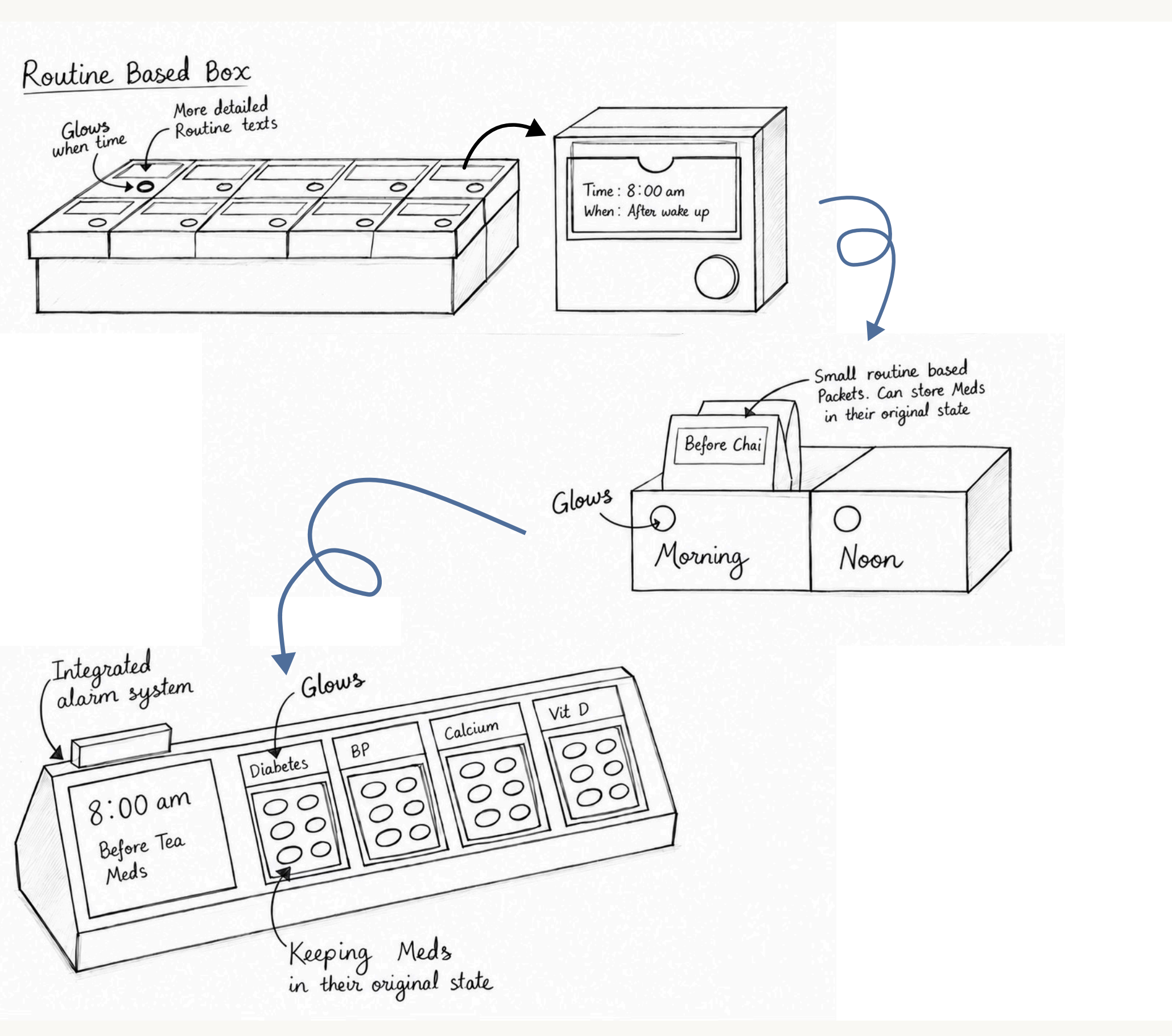
"Trusted by families" adds reassurance while choosing a caregiver.

Short 2–3 word insights on cards make trust easier to scan

Call requested
You'll receive a call from our care team within 10–15 minutes.
Priya Sharma Assigned Senior Caregiver Direct
Requested Just now
Expected call Within 10–15 min
Please keep your phone nearby and unmuted
Go back Home
Back to Priya's profile

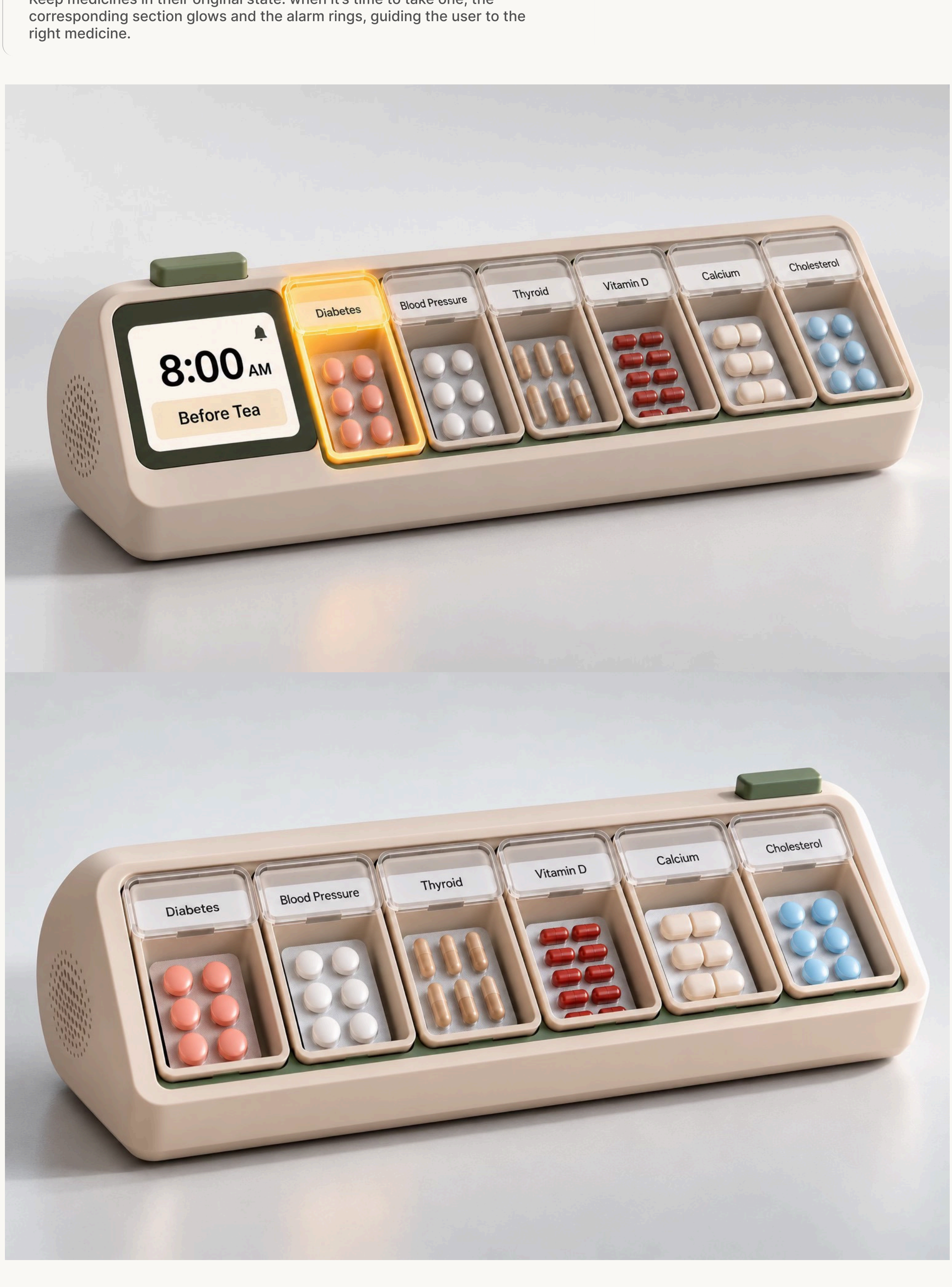
Pill box design

Pain point
users find it tedious to manually sort multiple medicines into day and night time based compartments, a repetitive task they have redo every week.



Pill box final mockup

Solution
Keep medicines in their original state. when it's time to take one, the corresponding section glows and the alarm rings, guiding the user to the right medicine.



Usability Testing

I tested the key flows to understand whether users could complete everyday medication and caregiving tasks easily and without confusion.

9 Participants
5 members -50+ yrs, 4 members-caregivers

6 Key flows tested

High-fidelity prototype

Moderated testing
Task-based + post-task interview

#	FLOW TESTED	AVG. TIME TAKEN	TASK COMPLETION	ERROR RATE	KEY ERRORS OBSERVED	KEY INSIGHT
1	**Add a medicine** Elderly app	1m 35s	80%	20%	Hesitation during medicine entry, difficulty selecting exact time; some needed help to proceed.	**Familiar routine choices made medicine setup easier than building a schedule manually.**
2	**Medicine reminder** Elderly app	18s	100%	7%	Users initially focused on the medicine rather than whether the associated routine had happened.	**Connecting reminders to everyday routines made the required action easier to understand.**
3	**Recognise & mark medicine taken** Elderly app	12s	100%	5%	Medicine name alone required more recall; some took longer to identify the correct medicine.	**Visual cues (image, colour, shape and purpose) improved medicine recognition.**
4	**Manage medicines** Caregiver app	25s	80%	20%	Confusion between stopping a reminder and permanently removing a medicine.	**A clear distinction between active and inactive medicines reduced confusion.**
5	**Add a medicine** Caregiver app	1m 10s	100%	10%	Most time spent configuring appearance and reminder details; uncertainty about mandatory fields.	**Breaking detailed medicine setup into smaller steps reduced complexity.**
6	**Track medication** Caregiver app	20s	100%	5%	Users searched for today's status first; broader information was needed only for longer-term tracking.	**Caregivers needed a quick daily status and a separate long-term adherence view.**
7	**Book a caregiver** Caregiver app	1m 10s	90%	10%	More time spent comparing caregivers than choosing date/time; users looked for experience, skills and trust cues.	**Trust and suitability mattered before users were ready to request care.**

CASE CLOSED.